Vinod Kumar

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PROFESSIONAL SUMMARY

Data-driven CSE graduate specializing in machine learning, proficient in Python, Java, SQL and core ML concepts with hands-on experience building dedicated predictive analytical projects. Highly disciplined professional with strong analytical skills developed during intensive full time civil services preparation, now seeking a foundational role as a fresher ML Engineer.

TECHNICAL SKILLS

Languages | Python, Java, SQL, HTML/CSS (Basic)
Libraries | pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn
Developer Tools | Git, GitHub, VS Code, Jupyter Notebook, MySQL

EDUCATION

B.Tech in Computer Science and Engineering| ABVGIET PRAGATINAGAR|2024

- CGPA: 8.21/10
- Relevant Coursework: Data Structure & Algorithms, Object-Oriented Programming, Database Management, Computer Networks, Theory of Computation, Applied Statistics

Full-Time Academic Preparation (UPSC)| Independent Study| June 2024-May 2026

- Dedicated 24 months to intensive study of advanced analytical reasoning, quantitative aptitude and data interpretation under strict timelines.
- Maintained technical continuity by independently engineering standalone machine learning projects and practicing clean coding conventions in Python.

PROJECTS

House Price Prediction System | Python, Linear Regression, Scikit-Learn, Pandas, NumPy, Matplotlib

- Developed an end-to-end machine learning regression pipeline to forecast residential property values based on key regional features.
- Engineered clean data pre-processing pipelines using Pandas to handle missing data values, scale numerical features, and apply one-hot encoding to categorical variables
- Evaluated model performance using Root Mean Squared Error (RMSE) and R-squared metrics, achieving a final predictive accuracy score of [96.8%].

Breast Cancer Prediction System| Python, Support Vector Classifier, Scikit-Learn, Pandas, NumPy, Matplotlib

- Developed a binary classification machine learning model to accurately predict and diagnose the nature of breast mass tumors as malignant or benign
- Implemented a Support Vector Classifier (SVC) to map clinical features into high-dimensional space and establish an optimal separating hyperplane.
- Evaluated classification performance using a confusion matrix, achieving a score [94%] Recall rate to minimize critical false-negative predictions.

E-Commerce Churn Predictor | Python, Random Forest, Scikit-Learn, MySQL, FastAPI

- Developed a **binary classification machine learning system** to predict whether an e-commerce customer is likely to churn based on behavioural and transactional features.
- Implemented a **Random Forest Classifier with RandomizedSearchCV** for hyperparameter optimization and evaluated model performance using accuracy and classification metrics.
- Built an **end-to-end ML application** integrating MySQL for data management, FastAPI for real-time predictions, and a web-based dashboard for customer risk analysis and prediction history logging.